THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 10: **Circles** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of tangents that can be drawn to a circle at a point on it is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** infinitely many

2. The length of the tangent from a point 17 cm from the centre of a circle of radius 8 cm is: 1

(a) 15 cm **(b)** 9 cm **(c)** 25 cm **(d)** 12 cm

3. A line that intersects a circle at two distinct points is called a: 1

(a) tangent **(b)** secant **(c)** radius **(d)** point of contact

4. The angle between a tangent and the radius at the point of contact is: 1

(a) 45° **(b)** 60° **(c)** 90° **(d)** 180°

5. A student states: “Two tangents can be drawn to a circle from a point inside it.” The correct response is: 1

(a) Correct

(b) Incorrect — no tangent can be drawn from an interior point

(c) Only one can be drawn

(d) Infinitely many can be drawn

6. TP and TQ are tangents from T to a circle with centre O. If $\angle PTQ = 70^\circ$, then $\angle POQ$ is: 1

(a) 70° **(b)** 110° **(c)** 90° **(d)** 140°

7. PQ is a tangent at P to a circle with centre O. $OP = 5$ cm and $OQ = 13$ cm. Student A says $PQ = 12$ cm; Student B says $PQ = 8$ cm ($= 13 - 5$). Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. Tangents PA and PB from P are inclined at 60° and $PA = 10$ cm. The length of chord AB is: 1

(a) 10 cm **(b)** 5 cm **(c)** $10\sqrt{3}$ cm **(d)** 20 cm

9. Tangents drawn at the ends of a diameter of a circle are: 1

(a) perpendicular **(b)** parallel **(c)** intersecting **(d)** coincident

10. A quadrilateral ABCD circumscribes a circle. If $AB = 6$ cm, $BC = 7$ cm and $CD = 4$ cm, then AD is: 1

(a) 3 cm **(b)** 5 cm **(c)** 4 cm **(d)** 9 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Prove that the tangents drawn at the ends of a diameter of a circle are parallel. 2

12. AB is a chord of a circle with centre O and AT is the tangent at A such that $\angle BAT = 60°$. Find $\angle AOB$. 2

13. TP and TQ are tangents from an external point T to a circle with centre O. Show that $\angle POQ + \angle PTQ = 180°$. 2

14. A student says that two tangents drawn from an external point to a circle are, in general, of unequal lengths. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that the lengths of the tangents drawn from an external point to a circle are equal. 3

16. Two concentric circles have radii 13 cm and 5 cm. Find the length of the chord of the larger circle which touches the smaller circle. 3

17. A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$. 3

OR

Prove that a parallelogram circumscribing a circle is a rhombus. 3

18. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$. 3

OR

A circle touches the sides BC, CA and AB of $\triangle ABC$ at D, E and F. If $AB = 11$ cm, $BC = 9$ cm and $CA = 8$ cm, find AF, BD and CE. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. Hence find the length of the tangent from a point 15 cm from the centre of a circle of radius 9 cm. 5

OR

A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC into which BC is divided by the point of contact D are 8 cm and 6 cm. Find the sides AB and AC. 5

20. **Case study:** A circular garden has centre O and radius 7 m. From a gate G, 25 m from O, two straight paths GA and GB touch the garden at A and B. 5

(i) Find GA.

(ii) Why is GA = GB?

(iii) Find $\angle OAG$.

(iv) Find the perimeter of the quadrilateral OAGB.

OR

Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre. Hence, if $\angle POQ = 125°$, find $\angle PTQ$. 5

END OF QUESTION PAPER — Total: 40 Marks

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